Masters Logbook

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Contents

1 Equations Of Motion

From Apostolos derivations we have:

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp\left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\text{SBP}}) = 0 \tag{1}$$

$$\ddot{\tilde{\chi}}_{\mathbf{k}} + 3H\dot{\tilde{\chi}}_{\mathbf{k}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle\right]\tilde{\chi}_{\mathbf{k}} = 0 \tag{2}$$

$$\langle\chi^2\rangle\ddot{\tilde{\theta}}_{\mathbf{k}} + (2\langle\chi\dot{\chi}\rangle + 3H\langle\chi^2\rangle)\dot{\tilde{\theta}}_{\mathbf{k}} + \frac{k^2}{a^2}\langle\chi^2\rangle\tilde{\theta}_{\mathbf{k}} = 0 \tag{3}$$

where the averages are

$$\langle\chi^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3}\left[|\tilde{\chi}_{\mathbf{p}}(t)|^2 + \text{2PI-corrections}\right]$$

$$\langle\dot{\theta}^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3}\left[|\dot{\tilde{\theta}}_{\mathbf{p}}(t)|^2 + \text{2PI-corrections}\right]$$

I want to rewrite these equations in a "vector" form for numerical integration, such that for instance

$$x = \tilde{\chi}_{\mathbf{k}} \quad \text{and} \quad v = \dot{\tilde{\chi}}_{\mathbf{k}}$$

so that

$$X = \begin{bmatrix} x \\ v \end{bmatrix} = \begin{bmatrix} \tilde{\chi}_{\mathbf{k}} \\ \dot{\tilde{\chi}}_{\mathbf{k}} \end{bmatrix}$$

Then

$$\dot{v} + 3Hv + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle\right]x = 0$$

To get a better understanding of these equations I will first compact some terms. Changing (1) into:

$$\ddot{\phi} + 3H\dot{\phi} - V(\phi, \langle\chi\rangle) = 0$$

with

$$V(\phi, \langle\chi^2\rangle) = \frac{\kappa V_0}{bM_{\text{Pl}}} \exp\left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\text{SBP}})$$

Then (2)

$$\ddot{\tilde{\chi}}_{\mathbf{k}} + 3H\dot{\tilde{\chi}}_{\mathbf{k}} + \omega\tilde{\chi}_{\mathbf{k}} = 0$$

$$\omega = \omega(\phi, \langle\chi^2\rangle, \langle\dot{\theta}\rangle, \langle(\nabla\theta)^2\rangle) = \frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle$$

and for (3) I am pretty sure I can just divide by $\langle\chi^2\rangle$

$$\ddot{\tilde{\theta}}_{\mathbf{k}} + \left(2\frac{\langle\chi\dot{\chi}\rangle}{\langle\chi^2\rangle} + 3H\right)\dot{\tilde{\theta}}_{\mathbf{k}} + \frac{k^2}{a^2}\tilde{\theta}_{\mathbf{k}} = 0$$

To make this managable computable, I think I need to first compute the averages of all.

2 Simpler equations

It makes sense to first start simulations with simpler equations where we take $\langle\dot{\theta}\rangle = \langle(\nabla\theta)^2\rangle = 0$ and forget about the third equation entirely.

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\mathrm{Pl}}}\exp\left[\frac{\phi}{bM_{\mathrm{Pl}}} - \kappa e^{\frac{\phi}{bM_{\mathrm{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\mathrm{SBP}}) = 0 \tag{4}$$

$$\ddot{\tilde{\chi}}_{\mathbf{k}} + 3H\dot{\tilde{\chi}}_{\mathbf{k}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\mathrm{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle\right]\tilde{\chi}_{\mathbf{k}} = 0 \tag{5}$$

where we can go a step further and first consider the homogeneous equation

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\mathrm{Pl}}}\exp\left[\frac{\phi}{bM_{\mathrm{Pl}}} - \kappa e^{\frac{\phi}{bM_{\mathrm{Pl}}}}\right] = 0 \tag{6}$$

To compute the averages we must discretise the integral over $\mathbf{k}$ modes, and first we will not consider any 2PI corrections

$$\langle\chi^2\rangle = \int_{\mathbb{R}^3}\frac{d^3p}{(2\pi)^3}|\tilde{\chi}_{\mathbf{p}}(t)|^2 \approx \sum_{\mathbf{p}}\omega_{\mathbf{p}}|\tilde{\chi}_{\mathbf{p}}(t)|^2$$

where $\omega_{\mathbf{p}}$ being the weights. There are several choices of $\omega_{\mathbf{p}}$, but since $\langle\chi^2\rangle$ is not isometric we use

$$\omega_{\mathbf{k}} = \frac{\delta k_x\delta k_y\delta k_z}{(2\pi)^3}$$

which is determined at the start of the simulation when we choose the change of k, assuming that the range is divided in equal steps that is $\delta k_x = \delta k_y = \delta k_z = \delta k \equiv \mathrm{const}$.

$$\omega_{\mathbf{k}} = \left(\frac{\delta k}{2\pi}\right)^3 = \omega$$

yielding the simplefied expression

$$\langle\chi^2\rangle \approx \omega\sum_{\mathbf{p}}|\tilde{\chi}_{\mathbf{p}}(t)|^2$$

Another important thing to note is that the hubble parameter H also changes, since it depends on the energy density ρ of the fields

$$\rho_\phi(t) = \frac{1}{2}\dot{\phi}^2 + V(\phi)$$
$$\rho_\chi(t) = \int\frac{d^3k}{(2\pi)^3}\frac{1}{2}(|\dot{\tilde{\chi}}_{\mathbf{k}}|^2 + \omega_k)$$

Then $\rho_{\mathrm{tot}} = \rho_\phi + \rho_\chi$, however at first we can neglect the ρ_χ term and the hubble parameter thus becomes

$$H \approx \sqrt{\frac{1}{3M_{\mathrm{pl}}^2}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right)} = \sqrt{\frac{1}{3M_{\mathrm{pl}}^2}\left(\frac{1}{2}\dot{\phi}^2 + V_0\exp\left[-\kappa e^{\frac{\phi}{bM_{\mathrm{Pl}}}}\right]\right)}$$

Note that we are using the potential $V(\phi) = V_0\exp\left[-\kappa e^{\frac{\phi}{bM_{\mathrm{Pl}}}}\right]$.

Next, I need to determine initial conditions. It makes sense to integrate from the symetry breaking point ϕ_{SBP}. During kination (point where energy density is dominated by kinetic energy) the ϕ field equation simplifies emensly:

$$\ddot{\phi} + 3H\dot{\phi} = 0$$

with a simplified Hubble parameter $H \approx \frac{|\dot{\phi}|}{\sqrt{6}M_{\text{pl}}}$. We can then solve these equations. Note that we will take $t_i = t_{SBP} = 0$ and $v = \dot{\phi}_{\text{SBP}}$

$$\phi = \sqrt{\frac{2}{3}}\text{sign}(v)\ln\left(1 + \sqrt{\frac{3}{2}\frac{|v|t}{M_{\text{pl}}}}\right)M_{\text{pl}}$$
$$\dot{\phi} = v\exp\left(-\sqrt{\frac{3}{2}}\frac{\phi}{M_{\text{pl}}\text{sign}(v)}\right)$$

That means that initial conditions $\phi_0 = \phi(t_i) = 0$ and $\dot{\phi}_0 = \dot{\phi}(0) = v$, where I think I can choose v - being a free parameter. We can similarly consider modes for χ field. For lange $|\phi|$ they satisfy

$$\ddot{\chi}_k + 3H\dot{\chi}_k + \omega_k^2\chi_k = 0$$

where $\omega^2 = k^2 + g^2\phi^2 - \lambda f^2$. The initial conditions are given by the Bunch-Davies vacuum state

$$\chi_{\phi,\text{vac}} = \frac{a^{-1}}{\sqrt{2\omega_k}}\exp\left\{-i\int^t \omega_k dt'\right\}$$

But again, since our initial conditions have $t = 0$ the above simplifies to just:

$$\chi_k(t_i) = \frac{1}{a(t_i)\sqrt{\omega_k}} = \frac{1}{\sqrt{\omega_k}}$$

Where we have used $a(t_i) = 1$. We can also differentiate χ_k modes to obtain initial conditions for $\dot{\chi}_k$, where we will only keep the adiabatic terms:

$$\dot{\chi} = [-H(t_i) - i\omega_k(t_i)]\chi_k(t_i) \approx 0$$

All of these initial conditions are dependent on parameters g, λ, f, v and technically t_i, though we set it to 0 previously. We can discuss the ranges of these parameters:

1. $10^{-5} < \lambda < 1$
2. $f = 5 \times 10^{-}7$
3. $10^{-14} < g < 1$
4. v is free

3 Integration strategy

First things first I need to change the variables from time to number of e-folds

$$t \to N, \quad \text{where} \quad N = \ln a$$

Then

$$\frac{dN}{dt} = \frac{\dot{a}}{a} = H \quad \Rightarrow \quad \frac{d\phi}{dt} = H\frac{d\phi}{dN} \quad \text{and} \quad \frac{d^2\phi}{dt^2} = H\frac{dH}{dN}\frac{d\phi}{dN} + H^2\frac{d^2\phi}{dN^2}$$

In prime notation for any function A

$$\dot{A} = HA' \quad \text{and} \quad \ddot{A} = H^2A'' + H'HA'$$

Then using Klein-Gordon, Friedman and acceleration equations

$$\begin{aligned}
&\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0 \\
&H^2 = \frac{1}{3M_{\rm pl}^2}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right) \\
&\dot{H} = -\frac{1}{2M_{\rm pl}^2}\dot{\phi}^2
\end{aligned}$$

We rewrite the derivatives

$$\begin{aligned}
&H^2\phi'' + H'H\phi' + 3H^2\phi' + V_{,\phi} = 0 \\
&H^2 = \frac{1}{3M_{\rm pl}^2}\left(\frac{1}{2}H^2\phi'^2 + V(\phi)\right) \\
&HH' = -\frac{1}{2M_{\rm pl}^2}H^2\phi'
\end{aligned}$$

The last two equations can be rewritten in a more convenient form: in the first I'm just expressing H^2 in the second I'm expressing H'/H, which are the terms that will appear when we rearrange KG

$$H^2 = \frac{V(\phi)}{3M_{\rm pl}^2 - \frac{1}{2}\phi'^2} \quad \text{and} \quad \frac{H'}{H} = -\frac{1}{2M_{\rm pl}^2}\phi'^2$$

Then after dividing KG by H^2 and inserting these relations I obtain

$$\phi'' + \left(3 - \frac{\phi'^2}{2M_{\rm pl}^2}\right)\phi' + \left(3M_{\rm pl}^2 - \frac{1}{2}\phi'^2\right)\frac{V_{,\phi}}{V} = 0$$

I'm not sure if I can do this, but I noticed that

$$W = \ln V \quad \Rightarrow \quad W_{,\phi} = \frac{V_{,\phi}}{V}$$

This is especially useful, since our potential is exponential, so using $V(\phi) = V_0 \exp\left[-\kappa e^{\frac{\phi}{bM_{\rm Pl}}}\right]$ we get

$$W = -\kappa \exp\left(\frac{\phi}{bM_{\rm pl}}\right) + W_0 \quad \text{and} \quad W_{,\phi} = -\frac{\kappa}{bM_{\rm pl}} \exp\left(\frac{\phi}{bM_{\rm pl}}\right)$$

where $W_0 = \ln V_0$.

For convenience I'll rewrite everything more explicitly with $M_{\rm pl} = 1$

$$\phi'' + \left(3 - \frac{\phi'^2}{2}\right)\phi' + \left(3 - \frac{1}{2}\phi'^2\right)\left(-\frac{\kappa}{b}\exp\left(\frac{\phi}{b}\right)\right) = 0$$

There is further massaging I can do, but for now I can also integrate this equation as is. Take $u_1 = \phi$ and $u_2 = \phi\prime$ then

$$\begin{aligned} u_1' &= u_2 \\ u_2' &= \left(\frac{1}{2}u_2^2 - 3\right)u_2 + \left(3 - \frac{1}{2}u_2^2\right)\frac{\kappa}{b}\exp\left(\frac{u_1}{b}\right) \end{aligned}$$

I can further masage this equation (Not sure if its worth to), by multiplying everything by ϕ', then $\phi''\phi' = \frac{1}{2}(\phi'^2)'$ and $W_{,\phi}\phi' = bM_{\rm pl}W'$, hence

$$\frac{1}{2}(\phi'^2)' + \left(3 - \frac{\phi'^2}{2M_{\rm pl}^2}\right)\phi'^2 + \left(3M_{\rm pl}^2 - \frac{1}{2}\phi'^2\right)bM_{\rm pl}W' = 0$$

Taking $M_{\rm pl} = 1$ and multiplying everything by 2

$$(\phi'^2)' + \left(6 - \phi'^2\right)\phi'^2 + \left(6 - \phi'^2\right)bW' = 0$$

Another approach, instead of multiplying everything by ϕ' we can just split these equations into a system with $u = \phi'$, then

$$\begin{cases} u = \phi' \\ u' + (6 - u^2)(u^2 + bW_{,\phi}) = 0 \end{cases}$$

Since I will be solving these equations numerically, I need some analytic results to check my calculations with. I will use slow-roll and kination approximations.

For slow-roll we consider that $|\ddot{\phi}| \ll |3H\dot{\phi}|$ and $|\ddot{\phi} \ll |V_{,\phi}(\phi)|$, so Friedman equation simplifies to

$$H^2 = \frac{V(\phi)}{3M_{\rm pl}^2}$$

And Slow-roll klein gordon (wrt time) equation becomes:

$$3H\dot{\phi} = -V_{,\phi}(\phi)$$

Again, using the same change of variables $t \to N$ we have

$$3H^2\phi' = -V_{,\phi} \quad \Rightarrow \quad \phi' = -M_{\rm pl}^2\frac{V_{,\phi}(\phi)}{V(\phi)} = -M_{\rm pl}^2 W_{,\phi}$$

This now should be solvable analytically, again, setting $M_{\rm pl} = 1$

$$N = \int -\frac{1}{W_{,\phi}} d\phi = \int \frac{b}{\kappa} \exp\left(-\frac{\phi}{b}\right) d\phi = -\frac{b^2}{\kappa} \exp\left(-\frac{\phi}{b}\right)$$

There is also the kination limit, which was discussed a bit earlier.

4 End of inflation

Note that inflation lasts until $\epsilon = 1$, where

$$\epsilon = -\frac{\dot{H}}{H^2} \quad \text{or} \quad \epsilon = \frac{\dot{\phi}^2}{2M_{\text{pl}}^2 H^2}$$

where for inflaton field we use $\rho_\phi = T^{00} = \frac{1}{2}\dot{\phi}^2 + V(\phi)$, assuming that the field is homogeneous, then $\nabla\phi = 0$. But this is not that necessary since we can change the integration coordinates into number of e-folds, also taking $\epsilon = 1$ and $M_{\text{pl}} = 1$, we get

$$\dot{\phi}^2 = 2H^2 \quad \Rightarrow \quad H^2(\phi')^2 = 2H^2 \quad \Rightarrow \quad \phi' = \pm\sqrt{2}$$

which means that at the end of inflation $\phi' = \sqrt{2}$ and in principle I can integrate backwards.

January

Notes from end of January

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I want to apply slowroll approximation to the inflaton evolution and only then change variables to the number of efolds. I had this solved numerically previously, but I think I accidentally removed it. Either way it's good to do it again and compare numerical calculations with analytic calculations.

In slow-roll regime kinetic energy is negligable compared to the potential and acceleration if negligible

$$\dot{\phi} \ll V(\phi) \quad \text{and} \quad |\ddot{\phi}| \ll |3H\dot{\phi}|$$

Then Friedmann equation simplifies to

$$H^2 = \frac{1}{3M_{\rm pl}^2}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right) \quad \rightarrow \quad H^2 = \frac{V(\phi)}{3M_{\rm pl}^2}$$

and the equation of motion for the inflaton simplifies to

$$3H\dot{\phi} + V_{,\phi}(\phi) = 0$$

Now switch integration variables.

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Switching the integration variable to the number of efolds

$$3H^2\phi' = -V_{,\phi} \quad \Rightarrow \quad \phi' = -\frac{V_{,\phi}(\phi)}{V(\phi)}$$

where we used $M_{\rm pl} = 1$. With $V(\phi) = V_0 \exp\left[-\kappa e^{\frac{\phi}{b}}\right]$ and $V(\phi)_{,\phi} = V(\phi)\frac{-\kappa}{b}\exp\left(\frac{\phi}{b}\right)$ we're left with

$$\phi' = \frac{\kappa}{b}\exp\left(\frac{\phi}{b}\right) \Rightarrow \int_{\phi_i}^{\phi_f} \frac{b}{\kappa}\exp\left(-\frac{\phi}{b}\right) d\phi = N_f - N_i = \Delta N$$

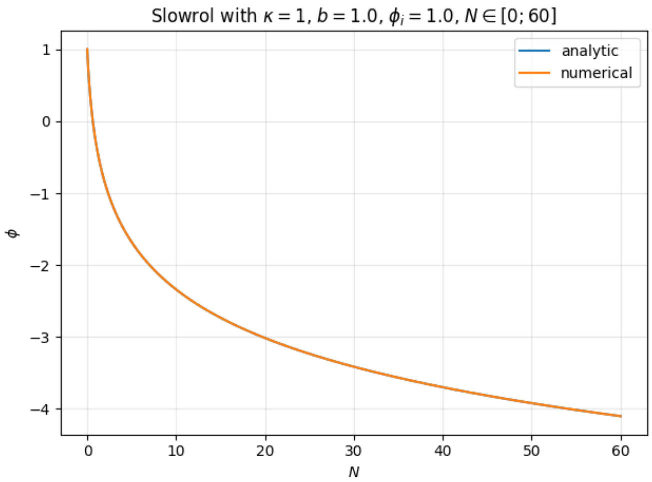
Integrating and inverting to obtain phi, we have

$$\Delta N = \frac{b^2}{\kappa}\left(e^{-\phi_i/b} - e^{-\phi_f/b}\right) \Rightarrow \phi_f = -b\ln\left(e^{-\phi_i/b} - \frac{\kappa}{b^2}\Delta N\right)$$

At the start of inflation, by definition $N_i = 0$ (if we define $N(t) \equiv \ln\frac{a(t)}{a(t_i)}$), we have then that slow-roll up until inflation end looks like:

$$\phi(N) = -b\ln\left(e^{-\phi_i/b} + \frac{\kappa}{b^2}N\right) \tag{7}$$

Implemented the numerical solution and compared with analytic solution:



They are so extremely close you can't even see the analytic solution. Next, I want to learn a bit about perturbations. I've found slides "Cosmic Inflation - Mini Lecture" by Jinsu Kim

They say that once the scalar potential $V(\phi)$ is given, then you can compute cosmological observables

- Calculate slow-roll parameters:

$$\epsilon = \frac{M_{\mathrm{pl}}^2}{2}\left(\frac{V_{,\phi}}{V}\right)^2, \qquad \text{and} \qquad \eta = M_{\mathrm{pl}}^2 \frac{V_{,\phi\phi}}{V}$$

- Find the end of inflation $\epsilon \approx 1$
- Find the horizon-crossing point $N = 60$ (i.e. CMB scale)

$$N = \int H dt, \qquad H^2 \approx \frac{V}{3M_{\mathrm{pl}}^2}$$

- Compute cosmological observables at the horizon-crossing point:

$$\mathcal{P}_s = \frac{V}{24\pi^2 M_{\mathrm{pl}}^4 \epsilon}, \quad n_s = 1 + 6\epsilon + 2\eta, \quad r = 16\epsilon$$

We know that there were perturbations at some point in the early universe due to the large scale structures we observe today, hence we need to spoil homogeneity and isotropy of the Universe. Assume that the inflaton field ϕ is decomposed as

$$\phi(t, \boldsymbol{x}) = \varphi(t) + \delta\phi(t, \boldsymbol{x})$$

where $\varphi(t)$ is the background field, only time dependent, $\delta\phi(t, \boldsymbol{x})$ is the perturbation (function of spacetime). The action is given by

$$S_{EH} = \frac{M_{\mathrm{pl}}^2}{2}\int d^4x\sqrt{-g}R,$$
$$S_\phi = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right] \tag{8}$$

with $S = S_{EH} + S_\phi$. The equations of motion for the inflaton is obtained by varying the acrtion wrt ϕ (we treat $g_{\mu\nu}$ as fixed).

$$\delta S_\phi = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\delta(\partial_\mu\phi\partial_\nu\phi) - V_{,\phi}(\phi)\delta\phi\right]$$

Then using the product rule and $\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)$

$$\delta(\partial_\mu\phi\partial_\nu\phi) = (\partial_\mu\delta\phi)\partial_\nu\phi + \partial_\mu\phi(\partial_\nu\delta\phi)$$

$$-\frac{1}{2}g^{\mu\nu}(\partial_\mu\delta\phi)\partial_\nu\phi + -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi(\partial_\nu\delta\phi)$$

but if we switch $\mu \leftrightarrow \nu$ on the second term and commute then we the same thing. Yielding

$$\delta S = \int d^4x\sqrt{-g}[-g^{\mu\nu}\partial_\mu\phi\partial_\nu(\delta\phi) - V']$$

As is usual in these sort of variations we integrate by parts and the boundary term vanished. In order to have the correct integration measure we also factor out $\sqrt{-g}$ hence

$$\delta S = \int d^4\delta\phi[\partial_\nu(\sqrt{-g}g^{\mu\nu}\delta_\mu\phi) - \sqrt{-g}V_{,\phi}(\phi)] = \int d^4\sqrt{-g}\delta\phi\left[\frac{1}{\sqrt{-g}}\partial_\nu(\sqrt{-g}g^{\mu\nu}\delta_\mu\phi) - \sqrt{-g}V_{,\phi}(\phi)\right]$$

and finally $\delta S = 0$ for arbitrary $\delta\phi$, hence we obtain the Euler-Lagrange equation and in principle we are done:

$$\frac{1}{\sqrt{-g}}\partial_\nu(\sqrt{-g}g^{\mu\nu}\delta_\mu\phi) - \sqrt{-g}V_{,\phi}(\phi) = 0.$$

However, we can remember that the divergence of a vector $\nabla_\mu V^\mu = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}V^\mu)$, but $\nabla^\mu\phi = \partial^\mu\phi = g^{\mu\nu}\partial_\nu\phi$ defines a vector, hence

$$\Box\phi \equiv \nabla_\mu\nabla^\mu\phi = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi) \tag{9}$$

where the $\Box$ is the curved d'Alembertian. Leaving the Klein-Gordon equation

$$\Box\phi - V'(\phi) = 0.$$

Again, in principle we are done, however for cosmology we use FLRW metric with homogeneous $\phi(t)$ and note that FLRW assumes isotropy hence we use spherical coordinates

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\varphi^2)\right],$$

$$g_{\mu\nu} = \mathrm{diag}\left(-1, \frac{a^2}{1-kr^2}, a^2r^2, a^2r^2\sin^2\theta\right),$$

$$g = \det(g_{\mu\nu}) = (-1)\cdot\frac{a^2}{1-kr^2}\cdot a^2r^2\cdot a^2r^2\sin^2\theta = -a^6r^4\sin^2\theta.$$

This now allows us to compute $\Box$, using equation (9)

$$\begin{aligned}
\Box\phi &= \frac{1}{\sqrt{a^6 r^4 \sin^2\theta}}\partial_\mu(\sqrt{a^6 r^4 \sin^2\theta}g^{\mu t}\partial_t\phi) \\
&= \frac{1}{a^3}\partial t(a^3\dot{\varphi}) \\
&= \frac{1}{a^3}[3a^2\dot{a}\dot{\varphi} + a^3\ddot{\varphi}] \\
&= \ddot{\varphi} + 3H\dot{\varphi}
\end{aligned}$$

where $H = \frac{\dot{a}}{a}$ is the hubble parameter. Finally giving the Klein-Gordon equation that we know

$$\boxed{\ddot{\varphi} + 3H\dot{\varphi} + V'(\varphi) = 0}$$

Next we take a general expression for the energy-momentum tensor

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_\phi}{\delta g^{\mu\nu}}$$

Note that to compute a general functional derivative $\frac{\delta S}{\delta X}$ we use

$$\delta S = \int d^4x \frac{\delta S}{\delta X(x)}\delta X(x).$$

Notice that there is no integration measure $\sqrt{-g}$! Roughly speaking, the energy momentum tensor is just a special case with the functional derivative being wrt the metric. So we again vary the action (8), but this time wrt the metric.

$$\begin{aligned}
\delta S_\phi &= \int dx^4 \sqrt{-g}\delta(\sqrt{-g})\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right] + \sqrt{-g}\left[-\frac{1}{2}\delta(g^{\mu\nu})\partial_\mu\phi\partial_\nu\phi\right] \\
&= \int dx^4 \left\{\left(-\frac{1}{2}\sqrt{-g}g_{\mu\nu}\right)\left[-\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - V(\phi)\right] + \sqrt{-g}\left[-\frac{1}{2}\partial_\mu\phi\partial_\nu\phi\right]\right\}\delta g^{\mu\nu}
\end{aligned}$$

We identify the integrand as $\delta S_\phi/\delta g^{\mu\nu}$ and the energy-momentum tensor is then

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\phi)\right]$$

The energy density $\rho = T^0_0$ and pressure $p\delta^i_j = T^i_j$ are

$$\rho = \frac{1}{2}\dot{\varphi}^2 + V(\varphi), \qquad p = \frac{1}{2}\dot{\varphi}^2 - V(\varphi).$$

This is all nice, but let's move on to perturbations!!! Cosmological perturbations are cool since the large scale structure of our Universe can also be accounted for by inflation. Remember how we decomposed ϕ, likewise we can decompose the spacetime metric $g_{\mu\nu}$ as its background homogeneous and isotropic FLRW metric $g^(0)_{\mu\nu}(t)$, which is only time dependent, and the perturbation part $\delta g_{\mu\nu}$, which is spacetime dependent:

$$\begin{aligned}
\phi(x) &= \varphi(t) + \delta\phi(x) \\
g_{\mu\nu} &= g^{(0)}_{\mu\nu}(t) + \delta g_{\mu\nu}(x)
\end{aligned}$$

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Had a meeting today with Mindaugas, to determine the initial value for ϕ_i we basically can pick any N since it's relative and a free variable, like time. There are basically three stages. First we determine the initial values analytically, integrate until ϕ_{SBP} (which we ourselves pick) and then go a bit back and integrate with all the χ and θ fields. Also instead of using $b = \sqrt{\alpha}$ I should just use α. Another useful graph to make would be $\phi - \dot{\phi}$ graph.

Again, we look at slowroll. Using ϵ

$$\epsilon = \frac{1}{2}\left(\frac{V_{,\phi}}{V}\right)^2 = \frac{1}{2}\left(\frac{\kappa}{b}\right)^2 \exp\left(\frac{2\phi}{b}\right),$$

we know that at the end of inflation $\epsilon = 1$, hence we can determine the value of ϕ_{end}

$$\phi_{\text{end}} = \frac{b}{2}\ln\left(\frac{2b^2}{\kappa^2}\right)$$

then from the derived slowroll formula (7) assuming inflation ends at N_{end} we have

$$\begin{aligned}
\phi_{\text{end}} &= -b\ln\left(e^{-\phi_i/b} + \frac{\kappa}{b^2}N_{\text{end}}\right)\\
\Rightarrow e^{-\phi_{\text{end}}/b} &= e^{-\phi_i/b} + \frac{\kappa}{b^2}N_{\text{end}}\\
\Rightarrow \phi_i &= -b\ln\left(e^{-\phi_{\text{end}}/b} + \frac{\kappa}{b^2}N_{\text{end}}\right) = -b\ln\left(\frac{\kappa}{\sqrt{2}b} - \frac{\kappa}{b^2}N_{\text{end}}\right).
\end{aligned} \tag{10}$$

But notice that this puts a constraint on the possible values for N_{end} or b

$$\frac{\kappa}{b^2}N_{\text{end}} < \frac{\kappa}{\sqrt{2}b} \quad \Rightarrow \quad N_{\text{end}} < \frac{b}{\sqrt{2}} \quad \text{or} \quad b > \sqrt{2}N_{\text{end}}.$$

To find the initial value for ϕ' we first find the derivative and then similarly compute ϕ'_{end}

$$\phi'(N) = -\frac{\kappa}{b}\frac{1}{e^{-\phi_i/b} + \frac{\kappa}{b^2}N}$$

we can do a little trick by noticing that the denominator contains the same terms as the equation (7) hence using (10) we have

$$\phi'(N) = -\frac{\kappa}{b}\frac{1}{e^{\phi(N)/b}}$$

where I intentionally left the exponential in the denominator, for later convenience. Finally at $N = 0$ we have $\phi(N = 0) = \phi_i$, whose value we already determinmed. Plugging it in and cleaning up we're left with

$$\phi'(0) = \frac{1}{\frac{N_{\text{end}}}{b} - \frac{1}{\sqrt{2}}}$$

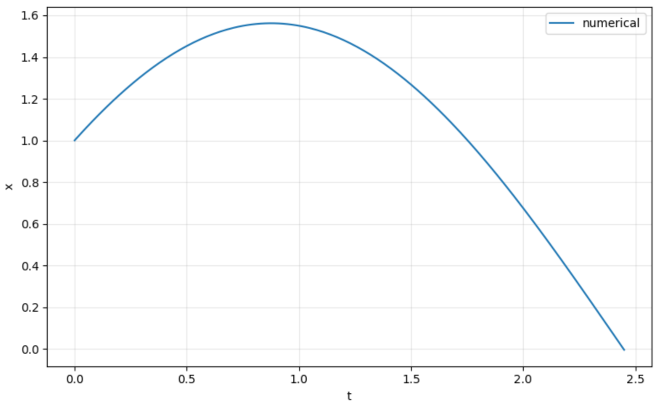
Now that we have all of the inital conditions we're able to integrate!!! For convenience here are all of the important formulas with $b = \sqrt{\alpha}$ inserted

$$\phi(0) = -\sqrt{\alpha}\ln\left(\frac{\kappa}{\sqrt{2\alpha}} - \frac{\kappa}{\alpha}N_{\text{end}}\right) \quad \text{and} \quad \phi'(0) = \frac{1}{\frac{N_{\text{end}}}{\sqrt{\alpha}} - \frac{1}{\sqrt{2}}}$$

February

02 02

Managed to create conditional integrator. The problem is that you don't know how long the integration might last so you can't allocate space for an array beforehand. There are basically a few solutions. You can append to numpy array, but every time you do that numpy has to reallocate the memory and its $\mathcal{O}(n^2)$ so super expensive. Another solution is to append to a regular array, which is $\mathcal{O}(1)$ and then you can allocate some memory beforehand, and if it exceedes the cap you can double itk, which is the solution I chose. First integrated a sample harmonic oscillator with a condition that if position reaches $x = x_{\text{end}} = 0$ the integration stops.



Been learning how to derive geodesics in GR. In general we can use the arc length action:

$$S[x^\mu(\lambda)] = -mc \int ds = -mc \int d\lambda \sqrt{-g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu},$$

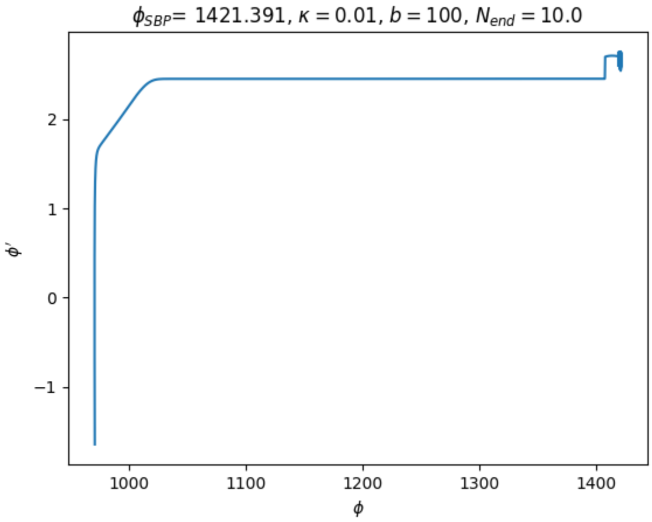
where $\dot{x}^\mu \equiv \frac{d^\mu}{d\lambda}$ and λ is a parameter along the worldline. To obtain the geodesics we extremise the action as usual. We can use a simpler and equivalent lagrangian

$$S'(x(\lambda)) = \frac{1}{2} \int d\lambda \, g_{\mu\nu}(x)\dot{x}^\mu \dot{x}^\nu$$

Now we can apply Euler-Lagrange equations:

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^\rho}\right) - \frac{\partial L}{\partial x^\rho} = 0$$

02 03



02 04

Repeated the calculations for initial values for ϕ and ϕ'. Reminder, we're using $V(\phi) = V_0 \exp\left(-\kappa e^{\frac{\phi}{b}}\right)$ and we're taking $M_{\text{pl}} = 1$. Start with the first slow-roll parameter ϵ

$$\epsilon = \frac{1}{2}\left(\frac{V_{,\phi}}{V}\right) \quad \Rightarrow \quad \epsilon = \frac{1}{2}\left(\frac{\kappa}{b}\right)^2 \exp\left(\frac{2\phi}{b}\right)$$

The inflation ends when $\epsilon = 1$ and this gives us ϕ_{end}

$$\phi_{\text{end}} = \frac{b}{2}\ln\left[\frac{2b^2}{\kappa}\right] \quad \Leftrightarrow \quad \exp\left[\frac{\phi_{\text{end}}}{b}\right] = \frac{\sqrt{2}b}{\kappa}$$

Using the slow-roll condition we can also obtain $\dot{\phi}_{\text{end}}$ by evaluating the slow-roll equation at ϕ_{end}

$$3H\dot{\phi} \approx -V_{,\phi}$$

Note that we don't really care for $\dot{\phi}_{\text{end}}$, at least for now. Using

$$H = \sqrt{\frac{V_0}{3}}\exp\left(-\sqrt{\frac{\alpha}{2}}\right) \quad \text{and} \quad V_{,\phi}(\phi) = V_0 \exp\left(-\kappa e^{\frac{\phi}{b}}\right)\left(\frac{-\kappa}{b}\right)e^{\frac{\phi}{b}}$$

then evaluating everything at ϕ_{end} and plugging into the slow-roll equation

$$\dot{\phi}_{\text{end}} = -\frac{V_{,\phi}}{3H} = -\frac{V_0 \exp\left(-\sqrt{2}b\right)\sqrt{2}}{3\frac{\sqrt{V_0}}{\sqrt{3}}\exp\left(-\frac{b}{\sqrt{2}}\right)} = \sqrt{\frac{2}{3}}\sqrt{V_0}\exp\left(-\frac{\sqrt{2}}{2}\sqrt{2}\right) = \sqrt{\frac{2}{3}V_{\text{end}}}$$

Next we solve the slow-roll equation and we can obtain the initial value ϕ_i

$$\phi' \equiv \frac{d\phi}{dN} = -\frac{V_{,\phi}(\phi)}{V(\phi)} \quad \Rightarrow \quad \frac{d\phi}{dN} = \frac{\kappa}{b}\exp\left(\frac{\phi}{b}\right)$$

$$\Rightarrow \int_{\phi_i}^{\phi_f} \frac{b}{\kappa}\exp\left(-\frac{\phi}{b}\right)d\phi = \int_{N_i}^{N_f} dN = N_f - N_i.$$

Note that we define $N \equiv \frac{\ln a}{\ln a(t_i)}$, then $N_i = 0$. Then

$$-\frac{b^2}{\kappa}\left[\exp\left(-\frac{\phi_f}{b}\right) - \exp\left(-\frac{\phi_i}{b}\right)\right] = N_f \tag{11}$$

At the end of inflation $\phi_f = \phi_{\text{end}}$

$$\frac{b^2}{\kappa}\left[\frac{\sqrt{2}\,b}{\kappa} - \exp\left(-\frac{\phi_i}{b}\right)\right] = -N_{\text{end}} \quad \Rightarrow \quad \left(\frac{b}{\sqrt{2}} + N_{\text{end}}\right) = \frac{b^2}{\kappa}\exp\left(-\frac{\phi_i}{b}\right) \tag{12}$$

$$\phi_i = -b\ln\left[\frac{\kappa}{b^2}\left(\frac{b}{\sqrt{2}} + N_{\text{end}}\right)\right]$$

And now we similarly like before can compute the initial ϕ_i' from the slow-roll equation

$$\phi' = \frac{-V_{,\phi}}{3H^2} \quad \Rightarrow \quad \phi' = \frac{\kappa}{b}\exp\left(\frac{\phi}{b}\right)$$

$$\phi_i' = \frac{b}{\frac{b}{\sqrt{2}} + N_{\text{end}}}$$

And so we get the correct expressions compared to Mindaugas paper. With a simple fix I was able to also produce some plots. First I varied κ from 100 to 300

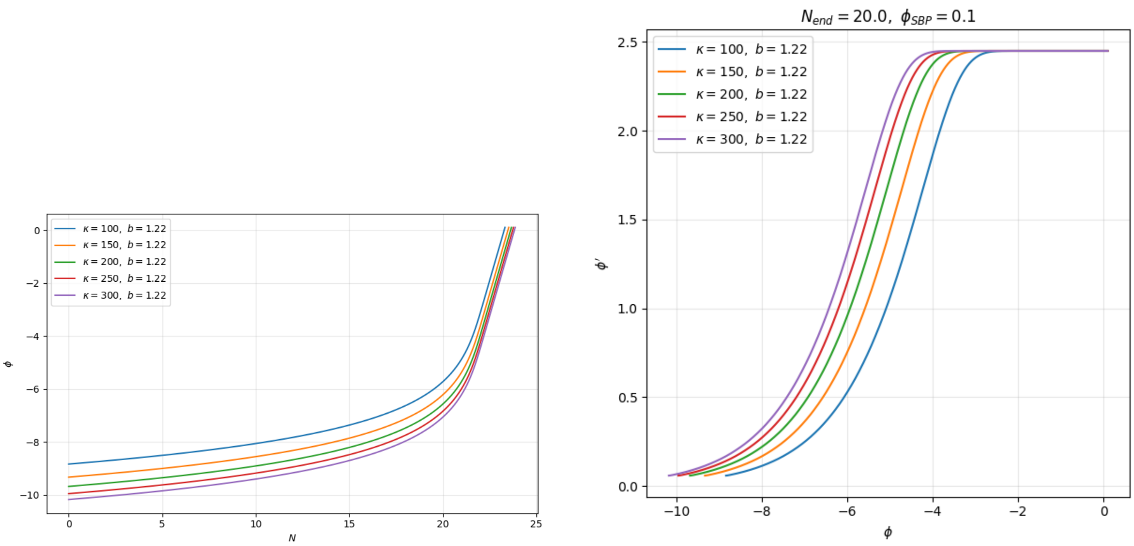


Figure 1: Varrying κ

Then I varied b (α implicitly)

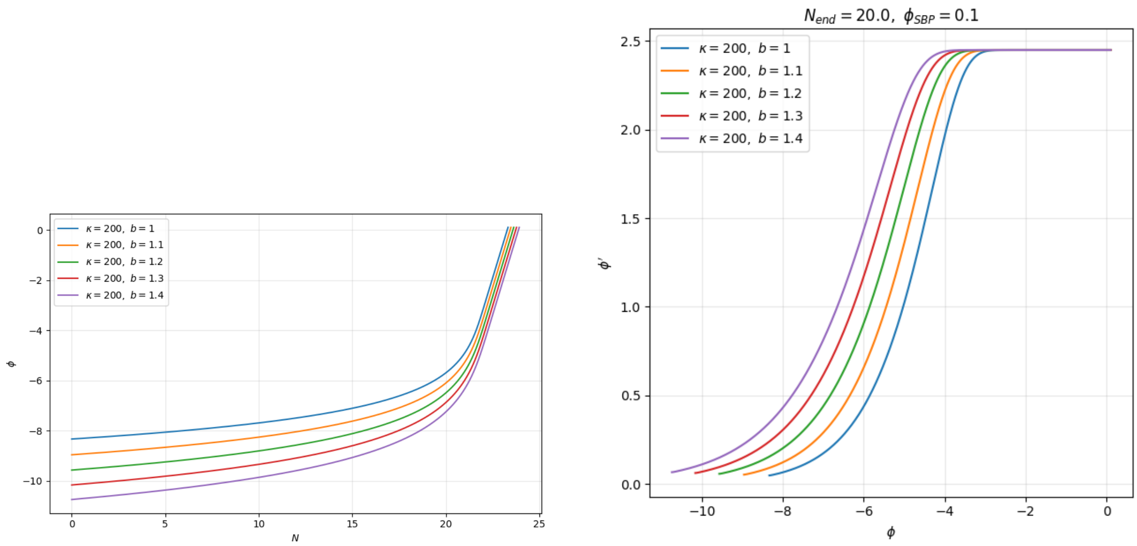


Figure 2: Varrying $\sqrt{\alpha}$

Tommorow I will probably try and compare the graphs with slowroll.

02 05

Due to previous mistakes we redired the formula for $\phi(N)$ we start from (11) then taking $N_f = N$ and exponential form of ϕ_i (12) to obtain

$$
\begin{aligned}
& -\frac{b^2}{\kappa}\left[\exp\left(-\frac{\phi(N)}{b}\right) - \exp\left(-\frac{\phi_i}{b}\right)\right] = N \\
\Rightarrow \quad & \exp\left(-\frac{\phi(N)}{b}\right) = -\frac{\kappa}{b^2}N + \exp\left(-\frac{\phi_i}{b}\right) \\
\Rightarrow \quad & \phi(N) = -b\ln\left[\frac{\kappa}{b^2}\left(N_{\text{end}} - N\right) + \frac{\kappa}{b\sqrt{2}}\right]
\end{aligned}
$$

Notice that $N \lesssim N_{\text{end}}$, due to the logarithm. Which just corresponds to the approximation working until the end of inflation, which makes sense. This now allows us to compare results

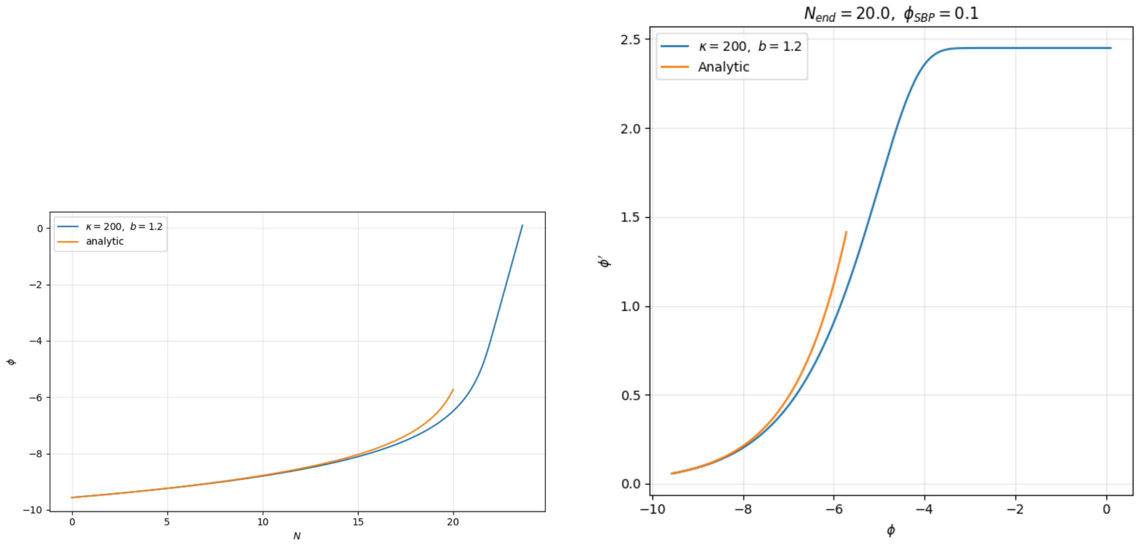


Figure 3: Numerical solutions vs Slow-roll analytic with $N_{\text{end}} = 20.0$, $\kappa = 200$, $b = \sqrt{\alpha} = 1.2$, $\phi_{SBP} = 0.1$

Next, I should probably explore kination. In the kination limit the kinetic energy dominates so $\dot{\phi} \gg V_(\phi)$. Then Klein-Gordon equation takes the form

$$\ddot{\phi} + 3H\dot{\phi} \simeq 0$$

Again, changing integration variables from time to efolds

$$H^2\phi'' + \dot{H}\phi' + 3H^2\phi' = 0 \quad \Rightarrow \quad \phi'' + \frac{\dot{H}}{H^2}\phi' + 3\phi' = 0$$

But in the kination limit the Friedman equation reduces to $H^2 = \frac{1}{3}\frac{1}{2}\dot{\phi}^2$. Hence

$$\frac{\dot{H}}{H^2} = \frac{-\frac{1}{2}\dot{\phi}^2}{\frac{1}{3}\frac{1}{2}\dot{\phi}^2} = -3.$$

Note, that this also gives the condition for kination in terms of ϵ, i.e. $\epsilon = 3$. Combining these two results we're left with a trivial equation

$$\phi'' - 3\phi' + 3\phi' = 0 \quad \Rightarrow \quad \phi'' = 0.$$

Which has the following solution

$$\phi(N) = \phi_0'(N - N_f) + \phi_0$$

Here ϕ_0 and ϕ_0' are initial conditions for kination. In the paper Mindaugas writes that these correspond to the end of simulation (we stop simulation when ϕ reaches ϕ_{SBP}), so we can take ϕ_0 from simulation and we can compute ϕ_0' with $\epsilon = 3$.

$$(\phi')^2 = 2\epsilon \quad \Rightarrow \quad \phi' = \sqrt{6}$$

Then

$$\phi(N) = \sqrt{6}\,(N - N_{\text{end}}) + \phi_0$$

Where N_{end} is the efold number when the simulation ends. Then we are able to plot the results

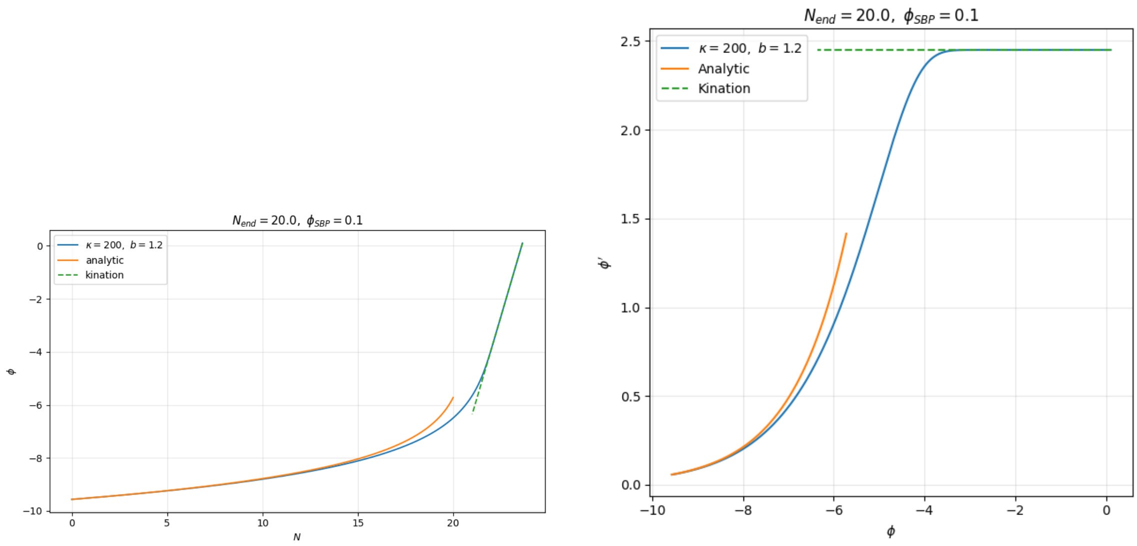


Figure 4: Numerical solutions vs Slow-roll analytic solution and kination

02 06

Now that we have the homogeneous part done, we can start simulating χ values. Again we take a simplified version at first with vanishing self-coupling $\lambda = 0$ and no evolution of θ. So from Apostolos equations

$$\ddot{\tilde{\chi}}_{\boldsymbol{k}} + 3H\dot{\tilde{\chi}}_{\boldsymbol{k}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2\right]\tilde{\chi}_{\boldsymbol{k}} = 0$$

I can try again to change the integration variable to the number of efolds. There is a lot of interesting content in the other paper also by Mindaugas.

Quitisential inflation is as I understand mostly defined by the shape of the potential: flat top to have slowroll, big gradient and then flat tail. Kination is a part of quitissential inflation and the fact that particles are not produced by the inflaton decay, but rather by non-perturbative effects.

02 07

Read part of the paper and got a slightly better understanding. One thing to note is that for initial conditions we take the Bunch-Davies vacuum state.

$$\chi_{k,\text{vac}} = \frac{a^{-1}}{\sqrt{2\omega_k}} e^{-i\int^t \omega_k dt'} \tag{13}$$

What are these Bunch Davies initial conditions? First we define conformal time

$$d\eta = \frac{dt}{a(t)} \quad \text{then} \quad ds^2 = a(\eta)^2(-d\eta^2 + d\boldsymbol{x}^2),$$

where the conformal time defines the causal distance (or reach), s.t. $\Delta\eta$ gives the maximum causal distance light could travel, and like in Minkowski space, the causality structure is identical. Conformal time makes spacetime conformally flat. Next, we will investigate the dynamics of a scalar field, in FLRW. Consider the scalar field $\phi(\eta, \boldsymbol{x})$ and the above FLRW metric. Then the action (just the kinetic part) is given by

$$S = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi\right].$$

Then from the metric we have $\sqrt{-g} = a^4(\eta)$, $g^{00} = -a^{-2}(\eta)$, $g^{ij} = \delta_{ij}a^{-2}(\eta)$. Inserting these into the metric we obtain

$$\begin{aligned} S &= \int d^4x\, a^4(\eta)\left[-\frac{1}{2}\left(\frac{1}{a^2}\dot{\phi}^2 + \frac{1}{a^2(\eta)}(\nabla\phi)^2\right)\right] \\ &= \int d^4x\, \frac{a^2}{2}\left(\phi'^2 - (\nabla\phi)^2\right). \end{aligned}$$

Varying the eqautions of motion with Euler-Lagrange equations (note that $\partial\mathcal{L}/\partial\phi = 0$) we're left with:

$$\ddot{\phi} + 2\frac{\dot{a}}{a}\dot{\phi} - \nabla^2\phi = 0.$$

Next we expand ϕ in fourier space

$$\phi(\eta, \boldsymbol{x}) = \int \frac{d^3k}{(2\pi)^3}\,\phi_{\boldsymbol{k}}(\eta)\,e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \quad \text{and} \quad \nabla^2\phi(\eta, \boldsymbol{x}) = -\int \frac{d^3k}{(2\pi)^3}\,k^2\,\phi_{\boldsymbol{k}}(\eta)\,e^{i\boldsymbol{k}\cdot\boldsymbol{x}}.$$

$$\begin{aligned} \int \frac{d^3k}{(2\pi)^3}\left[\ddot{\phi}_{\boldsymbol{k}}(\eta) + 2\frac{\dot{a}(\eta)}{a(\eta)}\dot{\phi}_{\boldsymbol{k}}(\eta) + k^2\phi_{\boldsymbol{k}}(\eta)\right]e^{i\boldsymbol{k}\cdot\boldsymbol{x}} &= 0 \\ \ddot{\phi}_{\boldsymbol{k}}(\eta) + 2\frac{\dot{a}(\eta)}{a(\eta)}\dot{\phi}_{\boldsymbol{k}}(\eta) + k^2\phi_{\boldsymbol{k}}(\eta) &= 0. \end{aligned}$$

Now we define a new variable $v(\eta, \boldsymbol{x}) = a(\eta)\phi(\eta, \boldsymbol{x})$, which in fourier space looks like $v_k(\eta) = a(\eta)\phi_k(\eta)$. Then we can express

$$\begin{aligned} \phi_k &= \frac{v_k}{a} \\ \dot{\phi}_k &= \frac{1}{a}\dot{v}_k - \frac{\dot{a}}{a^2}v_k \\ \ddot{\phi}_k &= \frac{1}{a}\ddot{v}_k - \frac{2\dot{a}}{a^2}\dot{v}_k + \left(\frac{2\dot{a}^2}{a^3} - \frac{\ddot{a}}{a^2}\right)v_k \end{aligned}$$

Inserting everything, combining terms and cleaning up

$$\frac{\ddot{v}_k}{a} - \frac{2\dot{a}}{a^2}\dot{v}_k + \left(\frac{2\dot{a}^2}{a^3} - \frac{\ddot{a}}{a^2}\right)v_k + \frac{2\dot{a}}{a^2}\dot{v}_k - \frac{2\dot{a}^2}{a^3}v_k + k^2\frac{v_k}{a} = 0 \quad \Rightarrow \quad \frac{\ddot{v}_k}{a} - \left(k^2 - \frac{\ddot{a}}{a}\right)\frac{v_k}{a} = 0$$

Which gives the final result:

$$\ddot{v}_k + \left(k^2 - \frac{\ddot{a}}{a}\right)v_k = 0 \tag{14}$$

For small wavelengths we can disregard $\ddot{a}/a$ which yields

$$v_k(\eta) \xrightarrow[k\eta\to-\infty]{} \frac{1}{\sqrt{2k}}\,e^{-ik\eta} \quad \text{Equivalently} \quad \phi_k(\eta) \xrightarrow[k\eta\to-\infty]{} \frac{1}{a(\eta)\sqrt{2k}}\,e^{-ik\eta}$$

where the normalization comes from the Wronskian normalization, which I don't really understand... Either way, more generally the equation (14) can be written

$$\ddot{v}_k + \omega_k^2(\eta)\,v_k = 0, \quad \text{where} \quad \omega_k^2(\eta) = k^2 + m_{\text{eff}}^2(\eta)$$

There is an idea to rewrite the equations for χ to have something like these, but that's maybe for the future.

I basically want to simulate

$$\ddot{\chi}_{\boldsymbol{k}} + 3H\dot{\chi}_{\boldsymbol{k}} + \omega_k\chi_{\boldsymbol{k}} = 0$$

where $\omega_k^2 = \frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2$. I am not sure which path to go, if I should change variables or not. I probably can try both. As for initial conditions, we need $\chi_k(t_0)$ and $\dot{\chi}_k(t_0)$ for the former we have equation (13) where the exponential just gives a phase, so I think I can just have a trivial phase of $e^0 = 1$ giving

$$\chi_k(t_0) = \frac{1}{a_0\sqrt{2\omega_{k,0}}} \tag{15}$$

To get the initial values for $\dot{\chi}(t_0)$ we can just take a derivative. Note that for now I will ignore the chi energy density in H. Otherwise, I would need to recombine fourier modes and I will leave that for later.

For now it makes sense to change the variables to e-folds as to have a simpler computation, since ϕ.

Rewriting the equation in terms of e-fold number gives

$$H^2\chi'' + HH'\chi' + 3H^2\chi' + \omega_k^2\chi = 0 \quad \text{or} \quad \chi'' + \left(\frac{H'}{H} + 3\right)\chi' + \frac{\omega_k^2}{H^2}\chi = 0$$

02 08

In *Kofman paper* I found a trick that can be easily derived, which sort of gets rid of the first derivative part. Suppose you have an equation of the form

$$\ddot{\chi} + 3\,\frac{\dot{a}}{a}\,\dot{\chi} + \omega^2\chi = 0.$$

Then we can introduce a new variable $\chi = a^n X$ and the goal is to find n such that the first derivative terms vanish and we're left if a harmonic-oscillator type of equation. The derivative terms give us

$$\dot{\chi} = na^{n-1}\dot{a}\,X + a^n\dot{X}$$
$$\ddot{\chi} = n(n-1)a^{n-2}\dot{a}^2 X + na^{n-1}\ddot{a}\,X + na^{n-1}\dot{a}\,\dot{\chi} + na^{n-1}\dot{a}\,\dot{X} + a^n\ddot{X}$$

Inserting into the equation and only looking at the first derivative terms we have

$$\left[2na^{n-1}\dot{a} + a^n 3\frac{\dot{a}}{a}\right]\dot{X}$$

We want these terms to vanish, so we set the coefficient to 0 and get $n = -\frac{3}{2}$. Then we can recombine the remaining terms, using the new value of n. Giving us just

$$\ddot{X} + \left[\omega^2 - \frac{3}{2}\frac{\ddot{a}}{a} - \frac{3}{4}\frac{\dot{a}^2}{a^2}\right]X = 0 \tag{16}$$

and by definition of the Hubble parameter $H = \frac{\dot{a}}{a}$ and $\dot{H} = \frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2}$ we can rewrite these equations even more concisely. This is just an illustrative example, maybe I can try and rewrite the equations of motion like that, but using number of efolds as my integration variable.

I tried that, but it doesn't work due to the additional +3 term. For now I will integrate wrt efold number, forget about optimisation and will discuss these things with Mindaugas. It can also be seen that it doesn't work from the above equation, if you were top introduce integration wrt efold number then the second derivative would give you first derivative.

02 09

I want to simulate these equations

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0,$$
$$\ddot{\chi}_k + 3H\dot{\chi} + \omega_k^2\chi = 0,$$

which in conformal time η take the form

$$\phi'' + 2\mathcal{H}\phi' + a^2 V_{,\phi} = 0, \tag{17}$$
$$\chi'' + 2\mathcal{H}\chi_k' + a^2\omega_k^2\chi_k = 0. \tag{18}$$

Here $\mathcal{H}$ is the hubble parameter in conformal time s.t. $\mathcal{H} = \frac{a'}{a}$. Then making substitutions $\Phi = a\phi$ and $X_k = a\chi_k$ we obtain equations, which are similar to Mukhanov equations

$$\Phi'' + \left(a^2 V_{,\phi\phi} - \frac{a''}{a}\right)\Phi = 0, \tag{19}$$
$$X_k'' + \left(a^2\omega_k^2 - \frac{a''}{a}\right)X_k = 0. \tag{20}$$

Don't know if there is much use out of these equations, but here they are.

Let's return to initla conditions for χ and $\dot{\chi}$. We already know the initial condition for χ that is equation (15). For $\dot{\chi}_k$ we first find the derivative and then again insert t_0.

$$\dot{\chi}_k = \frac{d}{dt}\left(\frac{a^{-1}}{\sqrt{2\omega_k}}\right)\exp\left(-i\int^t \omega_k dt'\right) + \frac{a^{-1}}{\sqrt{2\omega_k}}\frac{d}{dt}\left(\exp\left(-i\int^t \omega_k dt'\right)\right)$$

02 10

How do we determine the range for k modes? For large k, (UV bound) $\frac{k^2}{a^2} \gg H^2$ and so the dynamics becode that of a free wave $\chi \sim e^{\pm ikt}$. Similarly, for small k we have $\frac{k^2}{a^2} \ll H^2$, which entails that ω_k^2 becomes unimportant. But note that the bounds move with time $\frac{k}{aH} \propto e^{-N}$ and logarithmic spacing is required.

$$k_{\min} = a(t_{\text{start}})H(t_{t_{start}}), \qquad k_{\min} = \zeta a(t_{\text{SBP}})H(t_{t_{SBP}})$$

with $\zeta \approx 10 - 100$.

02 11

It does seem like at least for now it is the most useful to do the computation for homogeneous ϕ field wrt the number of efolds. The problem that I encountered was converting from N to t. We use the simple relation

$$\dot{\phi} = H\phi' \quad \Rightarrow \quad \frac{d\phi}{dt} = H\frac{d\phi}{dN} \quad \Rightarrow \quad \int dt = \int \frac{1}{H} dN$$

and we determine H via Friedman equation in N

$$H^2 = \frac{V(\phi)}{3 - \frac{1}{2}\phi'^2}.$$

In code the integration is made with the trapezoid rule.

02 12

Wrote the code for converting N to t and plotted the results

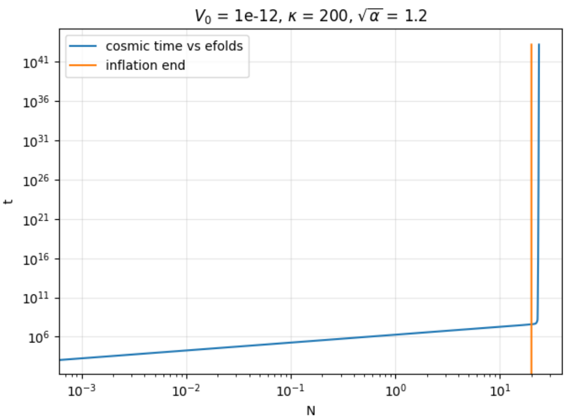


Figure 5: The orange vertical line marks $N = 20$, i.e. the end of inflation. Both N and t are in log scales.

To get a better view of the $N(t)$ after the inflation it's usefull to "zoom in" and "turn off" log scales. Note that to find the intersection we just check when the difference of N and $N_{\text{end}} = 20$ changes sign.

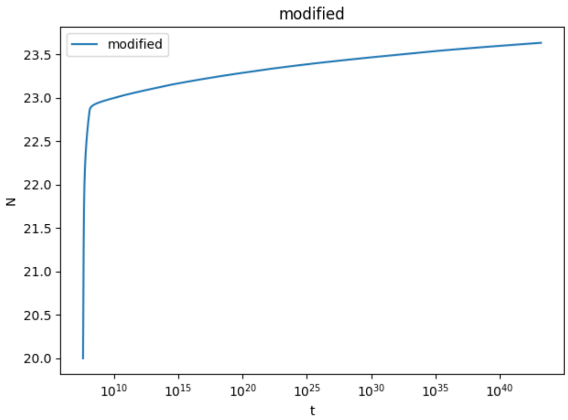


Figure 6: t is on log scale, while N is linear

For missing values one can interpolate.

Initial conditions yet again. One needs to be careful when looking at the initial conditions from Mindaugas papaer, since our $\omega_k^2 = \frac{k^2}{a^2} + \dots$ i.e. we use additional division by a^2. To start, consider the WKB approximation. As a reminder, consider the differential equation

$$y''(x) + Q(x)y(x) = 0 \quad \text{or} \quad y''(x) + \omega^2(x)y(x) = 0$$

where $\omega(x) = \sqrt{Q(x)}$. WKB approximation is built on the assumption that $\omega(x)$ varies slowly. To put that mathematically, if $\omega(x) = \text{const}$ then the solution would be $y(x) \sim e^{\pm i\omega x}$ with a natural wavelength $\lambda(x) = \frac{2\pi}{\omega(x)}$, meaning $\Delta x \sim \frac{1}{\omega}$. And from Taylor series we have $\Delta\omega \simeq \omega'(x)\Delta x$, thus

$$\Delta x \sim \frac{1}{\omega} \quad \Rightarrow \quad \Delta\omega \sim \frac{\omega'}{\omega}$$

But we want for ω to change very little during one oscillation, meaning $\frac{\Delta\omega}{\omega} \ll 1$. Hence the condition $\left|\frac{\omega'}{\omega^2}\right| \ll 1$. To use WKB approximation we use the ansatz $y = A(x)e^{iS(x)}$, then

$$\begin{aligned} y' &= (A' + iAS')e^{iS} \\ y'' &= (A'' + 2iA'S' + iAS'' - AS'^2)e^{iS} \end{aligned}$$

then

$$A'' + 2iA'S' + iAS'' - AS'^2 + \omega^2 A = 0$$

Let's now take a step back. Again, is ω was constant the exponential factor would be $\omega_k x$, which in all actuallity is the integral, so we have $S' \sim \omega$ (writting the derivative so that we can easily see the ω), then $S'' \sim \omega'$. Now we can investigate the asymptotic behaviour. Split the equation into real and imaginary parts. The imaginary part gives:

$$2A'S' + AS'' = 0 \quad \Rightarrow \quad \frac{A'}{A} = -\frac{S''}{2S'} \quad \Rightarrow \quad \frac{A'}{A} \sim \frac{\omega'}{\omega}.$$

Likewise, the real part gives

$$A'' - AS'^2 + \omega^2 A = 0 \quad \Rightarrow \quad A'' = (S'^2 - \omega^2)A \quad \Rightarrow \quad A'' \sim (\omega^2 - \omega^2)A = 0$$

which means that A'' is negligable. This now simplifies the main equation to

$$-AS'^2 + \omega^2 A + 2iA'S' + iAS'' = 0$$

After seperating real and imaginary parts we obtain:

$$-AS'^2 + \omega^2 A = 0 \quad \text{and} \quad 2A'S' + AS'' = 0$$

which yield the solutions $S(x) = \pm \int^x \omega(x')dx'$ and $A(x) = \frac{C}{\sqrt{\omega(x)}}$. Hence the WKB solution is

$$y(x) \simeq \frac{C}{\sqrt{\omega_k}} \exp\left(-i \int^t \omega_k(x')dx'\right) \tag{21}$$

Next we investigate the normalization with the Wronskian

02 13

Consider some complex KG field $\hat{\phi}(t, \boldsymbol{x})$

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left(\phi_{\mathbf{k}}(t)\, \hat{a}_{\mathbf{k}}\, e^{i\mathbf{k}\cdot\mathbf{x}} + \phi^*_{\mathbf{k}}(t)\, \hat{a}^\dagger_{\mathbf{k}}\, e^{-i\mathbf{k}\cdot\mathbf{x}}\right)$$

and conjugate momenta $\hat{\pi}(t.\boldsymbol{x}) = \dot{\hat{\phi}}(t, \boldsymbol{x})$

$$\hat{\pi}(t, \mathbf{x}) = \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left(\phi'_{\mathbf{q}}(t)\, \hat{a}_{\mathbf{q}}\, e^{i\mathbf{q}\cdot\mathbf{x}} + \phi^{*\prime}_{\mathbf{q}}(t)\, \hat{a}^\dagger_{\mathbf{q}}\, e^{-i\mathbf{q}\cdot\mathbf{x}}\right).$$

Thus from the canonical commutation relations $[\hat{\phi}(t, \boldsymbol{x}), \hat{\pi}(t, \boldsymbol{y})] = i\delta^3(\boldsymbol{x} - \boldsymbol{y})$ we have

$$\begin{aligned}
[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \Big(& \phi_{\mathbf{k}}(t)\, \phi'_{\mathbf{q}}(t)\, [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{q}}]\, e^{i(\mathbf{k}\cdot\mathbf{x} + \mathbf{q}\cdot\mathbf{y})} \\
& + \phi_{\mathbf{k}}(t)\, \phi^{*\prime}_{\mathbf{q}}(t)\, [\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{q}}]\, e^{i(\mathbf{k}\cdot\mathbf{x} - \mathbf{q}\cdot\mathbf{y})} \\
& + \phi^*_{\mathbf{k}}(t)\, \phi'_{\mathbf{q}}(t)\, [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}_{\mathbf{q}}]\, e^{i(-\mathbf{k}\cdot\mathbf{x} + \mathbf{q}\cdot\mathbf{y})} \\
& + \phi^*_{\mathbf{k}}(t)\, \phi^{*\prime}_{\mathbf{q}}(t)\, [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{q}}]\, e^{i(-\mathbf{k}\cdot\mathbf{x} - \mathbf{q}\cdot\mathbf{y})} \Big).
\end{aligned}$$

but $[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{q}}] = [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{q}}] = 0$ and $[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{q}}] = -[\hat{a}^\dagger_{\mathbf{k}}, \hat{a}_{\mathbf{q}}] = (2\pi)^3\delta^3(\boldsymbol{x} - \boldsymbol{y})$. Hence

$$\begin{aligned}
&= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \Big(\phi_{\mathbf{k}}(t)\, \phi^{*\prime}_{\mathbf{q}}(t)\, (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{q})\, e^{i(\mathbf{k}\cdot\mathbf{x} - \mathbf{q}\cdot\mathbf{y})} \\
&\qquad - \phi^*_{\mathbf{k}}(t)\, \phi'_{\mathbf{q}}(t)\, (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{q})\, e^{i(-\mathbf{k}\cdot\mathbf{x} + \mathbf{q}\cdot\mathbf{y})} \Big) \\
&= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left(\phi_{\mathbf{k}}(t)\, \phi^{*\prime}_{\mathbf{k}}(t)\, e^{i\mathbf{k}\cdot(\mathbf{x} - \mathbf{y})} - \phi^*_{\mathbf{k}}(t)\, \phi'_{\mathbf{k}}(t)\, e^{i\mathbf{k}\cdot(\mathbf{y} - \mathbf{x})} \right). \\
&= (\phi_{\mathbf{k}}(t)\, \phi^{*\prime}_{\mathbf{k}}(t) - \phi^*_{\mathbf{k}}(t)\, \phi'_{\mathbf{k}}(t))\, \delta^{(3)}(\mathbf{x} - \mathbf{y}).
\end{aligned}$$

Them comparing to the RHS of canonical commutation relation we have

$$\phi_{\mathbf{k}}\,\phi_{\mathbf{k}}^{*\prime} - \phi_{\mathbf{k}}^{*}\,\phi_{\mathbf{k}}' = i \tag{22}$$

Which is defined as the Wronskian W.

We now can combine everything. There are two ways of doing this, i.e. how we define the vacuum. We either define it from conformal time or from cosmic time. First, let's take a look at the latter

Consider the equation (16), and rewrite it as

$$\ddot{X} + \Omega_k^2 X = 0 \quad \text{with} \quad X = a^{3/2}\chi$$

where $\Omega_k^2(t) = \omega_k^2 - \frac{3}{2}\frac{\ddot{a}}{a} - \frac{9}{4}\frac{\dot{a}^2}{a^2} = \omega_k^2 - \frac{3}{2}\dot{H} - \frac{9}{4}H^2$. Then using WKB approximation we know the solution must look like equation (21). Hence

$$X_k = \frac{C}{\sqrt{2\Omega_k(t)}}\exp\left(-i\int^t \Omega_k(t')\,dt'\right) \quad \text{and} \quad X_k^* = \frac{C}{\sqrt{2\Omega_k(t)}}\exp\left(i\int^t \Omega_k(t')\,dt'\right)$$

$$\dot{X}_k = -\frac{1}{2}\frac{C\,\dot{\Omega}}{\sqrt{2\Omega}\,\sqrt{\Omega}}\exp\left(-i\int^t \Omega(t')\,dt'\right) - i\,\frac{C}{\sqrt{2\Omega_k(t)}}\exp\left(-i\int^t \Omega(t')\,dt'\right)\Omega(t)$$

$$\Rightarrow \dot{X}_k = -\frac{1}{2}\frac{\dot{\Omega}}{\Omega}X_k - i\Omega X_k \quad \text{and} \quad \dot{X}_k^* = -\frac{1}{2}\frac{\dot{\Omega}}{\Omega}X_k^* + i\Omega X_k^*$$

Then note that $X_u X_u^* = \frac{C^2}{2\Omega_u(t)}$ and by using the Wronskian (22) we obtain

$$X_k\dot{X}_k^* - X_k^*\dot{X}_k = i \quad \Rightarrow \quad 2i\,X_kX_k^*\Omega = i \quad \Rightarrow \quad 2\frac{C^2}{2\Omega(t)}\Omega = 1 \quad \Rightarrow \quad C = \pm 1$$

We only take the positive modes. Hence we obtain the vacuum solution

$$\chi_k = \frac{a^{-3/2}}{\sqrt{2\Omega_k}}\exp\left(-i\int^t \Omega_k(t')\,dt'\right) \quad \text{where} \quad \Omega_k^2(t) = \omega_k^2 - \frac{3}{2}\dot{H} - \frac{9}{4}H^2$$

There is an alternative way of computing the vacuum, namely by first converting to conformal time and only then following the same procedure. It is often considered a good practice to quantize everything in conformal time, since it maintains causal structure. The equations of motion would then be equations (17) and after substitution $X = a\chi$ we obtain Mukhanov type of equations (19). This now allows us to use the WKB approximation again.

$$X_k'' + \Omega_k^2(\eta)X_k = 0 \quad \text{where} \quad \Omega_k^2(\eta) = a^2\omega_k^2 - \frac{a''}{a}$$

After a similar procedure as before we obtain

$$\chi_k = \frac{a^{-1}}{\sqrt{2\Omega_k}}\exp\left(-i\int^\eta \Omega_k(\eta')d\eta'\right)$$

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The final step is to convert this expression to cosmic time, so that we could compare the vacuum states. First, remember that conformal time is defined as $d\eta = \frac{dt}{a}$, so

$$a' = \frac{d}{d\eta} = \frac{dt}{d\eta}\frac{d}{dt}(a) = a\dot{a} \quad \text{and} \quad a'' = a\frac{d}{dt}(a\dot{a}) = a(\dot{a}^2 + a\ddot{a})$$

then

$$\Omega_k^2(\eta) = a^2\omega_k^2 - \frac{a''}{a} = a^2\omega_k^2 - (\dot{a}^2 + a\ddot{a}) = a^2(\omega_k^2 - \dot{H} - 2H^2)$$

For convenience define $\Omega_k^2 = a^2\tilde{\Omega}_k^2$, so $\tilde{\Omega}_k^2 = \omega_k^2 - \dot{H} - 2H^2$. Then the exponential part becomes

$$\exp\left(\int^{\eta} \Omega_k(\eta')d\eta'\right) = \exp\left(\int^{\eta} \Omega_k(t')\frac{d\eta'}{a}\right) = \exp\left(\int^{t} \tilde{\Omega}_k dt'\right)$$

Combining everything we obtain

$$\chi_k = \frac{a^{-3/2}}{\sqrt{2\tilde{\Omega}_k}}\exp\left(\int^{t} \tilde{\Omega}_k dt'\right) \quad \text{where} \quad \tilde{\Omega}_k^2 = \omega_k^2 - \dot{H} - 2H^2$$

This is different from the previous result. Since we just want initial conditions for the numerical simulations at $t = 0$ we have $a = 1$ and a lot of stuff simplifies, leaving

$$\text{Cosmic Time:} \qquad \chi_k = \frac{1}{\sqrt{2\Omega_k}} \quad \text{with} \quad \Omega_k^2 = \omega_k^2 - \frac{3}{2}\dot{H} - \frac{9}{4}H^2$$

$$\text{Conformal Time:} \qquad \chi_k = \frac{1}{\sqrt{2\tilde{\Omega}_k}} \quad \text{with} \quad \tilde{\Omega}_k^2 = \omega_k^2 - \dot{H} - H^2$$

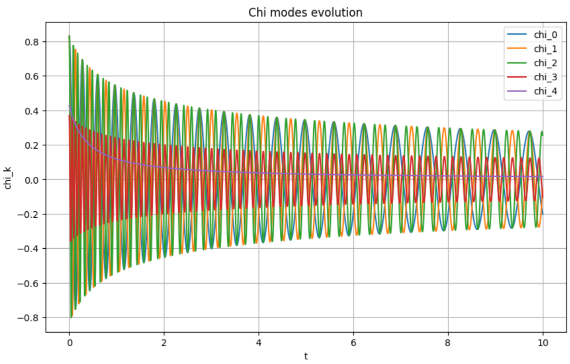
and for $\dot{\chi}$

$$\text{Cosmic Time:} \qquad \dot{\chi}_k = \left(-\frac{3}{2}H - \frac{1}{2}\frac{\dot{\Omega}_k}{\Omega_k} - i\,\Omega_k\right)\chi_k \quad \text{with} \quad \dot{\Omega}_k = \frac{1}{2\Omega_k}\left[\dot{\omega}_k^2 - \frac{3}{2}\ddot{H} - \frac{9}{2}H\dot{H}\right]$$

$$\text{Conformal Time:} \qquad \dot{\chi}_k = \left(-\frac{3}{2}H - \frac{1}{2}\frac{\dot{\tilde{\Omega}}_k}{\tilde{\Omega}_k} - i\,\tilde{\Omega}_k\right)\chi_k \quad \text{with} \quad \dot{\tilde{\Omega}}_k = \frac{1}{2\tilde{\Omega}_k}\left[\dot{\omega}_k^2 - \ddot{H} - 2H\dot{H}\right]$$

with $\dot{\omega}_k^2 = -2H\frac{k^2}{a^2} + 2g^2(\phi - \phi_{\mathrm{SBP}})\dot{\phi}$.

After simulating everything we obtain a preliminary plot


Chi modes evolution
chi_0
chi_1
chi_2
chi_3
chi_4
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
chi_k
0
2
4
6
8
10
t